

A Crash-Proof Team of Coding Agents

Claude Code remembers the conversation; Temporal remembers the job. Kill the worker mid-task and the agent finishes anyway: the same session, resumed from its last heartbeat rather than any saved checkpoint. First that makes one job crash-proof. Then many such jobs, each a single-purpose durable step with its own mandate, compose into a governed team, most of them never touching the agent at all.



There is a moment in every infrastructure demo where you stop trusting the slides and ask the presenter to pull the plug. We opened this project by pulling it ourselves. The setup was ordinary on purpose. A worker process was running a headless coding agent (headless meaning one command in a terminal, no chat window) on a small test-first Python task. The agent was minutes in, actively writing files, nothing finished, nothing saved anywhere. We killed the worker's entire process tree with an unblockable signal. No goodbye, no flush, no checkpoint.

For about two minutes, nothing happened at all. Then a different process, on a restarted worker, noticed the silence. It picked up the same half-finished conversation and ran it to completion: the same session, remembering everything it had learned. The total bill was four cents. Keep the four cents; the receipt shows up later. And the staged kill is only the legible version of what happens uninvited every week: an API overload, a rate limit, a deploy that restarts the worker.



The kill, as witnessed. No mechanism in the frame: it came back, and the how is where this article starts.

How that session survived is the first half of this article. The bigger half is what survival buys once you have it: a durable team. The work splits into single-purpose durable jobs. Only one kind of step runs Claude Code itself, and it does all the judgment work: building the feature, reviewing the diff, resolving the conflict. The rest carry the delivery pipeline around it, each job governed by playbooks it actually follows and rules re-asserted before every slice of work. That team carried a filed GitHub issue all the way to a merged pull request, resolving a real merge conflict along the way, with no human decision in the loop. One mechanical asterisk rides on that sentence, footnoted where the story is told.

*The fine print, up front: one engineer ran everything here, and the "we" is editorial. Every measured run used `haiku`, Claude's cheap fast tier, in July and August 2026; the run counts are single-digit, so every number is a point estimate that traces to a results file in the project's evidence archive. Dollar figures are haiku-priced; a stronger tier scales them up. And "crash-proof" here means process and worker crashes, the staged kill included; a machine that never comes back is an admitted gap, covered in *What This Doesn't Solve*.*

The Agent Remembers the Conversation

Start with the crime scene, and take inventory of what the kill failed to destroy. The first survivor is a file. Run Claude Code headless and it executes the whole agent, prints a result, and exits. The result carries a **session ID**: hand it back later with a resume flag and the agent reopens that exact conversation, because the transcript is an ordinary file on disk, filed under the directory the agent ran in. The *conversation* is already durable, and not as a toy mode: the official GitHub Action ships the same headless agent into continuous integration to review pull requests and turn issues into them.¹²

The second survivor is the guardrail harness, which matters the moment you hand an agent a shell: dangerous commands (deleting a tree, escalating privileges, pushing to a remote) are denied by the tool itself rather than discouraged in a prompt, hooks fire around every tool call, the operating system sandboxes what the process can touch, and a required output schema shapes what the agent must return.³ A line in a prompt is a suggestion; a deny rule is a fact, identical on the first attempt and the sixth.

Then the inventory comes up one item short. A resumed session is local to the machine it was born on, a limit the documentation itself concedes (newer CLI versions will find a session ID from any directory on that machine, but never from another one). There is no way to move it across machines, no lease, nothing that notices the machine died and carries the work elsewhere.⁴ The agent remembers the conversation. **Nothing remembers the job.**

A Durable Agent System = Three Parts

The Conversation: SHIPS
A Transcript on Disk,
Resumable by ID

The Guardrails: SHIP
Deny Rules · Hooks ·
OS Sandbox

The Job: EMPTY
The Part That Would Notice
a Dead Machine and
Move the Work

Half a Durable System. Two plates in, one socket empty. The color key, used across this family: indigo = the durable machinery · amber = judgment (agent or human) · purple = the durable record · green = a good exit · red = failure · dashed = crosses a team or a poll boundary.

Nothing Remembers the Job

The obvious way to drive a headless agent is a loop. Ask it to continue, refresh the session ID from the result, ask it to continue again:

```
while ;; do claude -p "continue" --resume "$SID" --max-turns 6; done
```

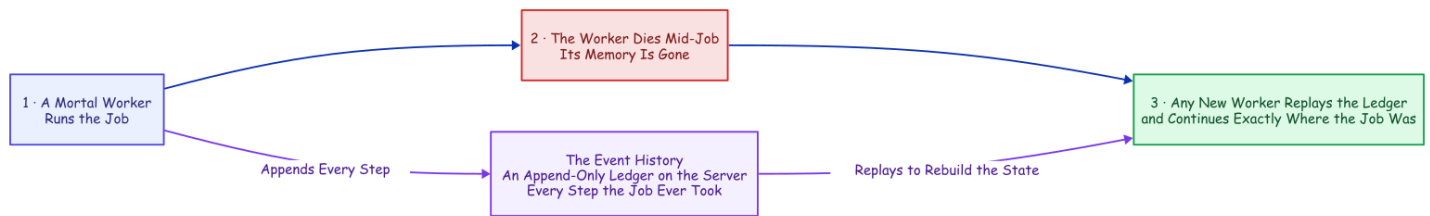
The real thing parses each run's output to keep the session ID current, but that is the shape. Two terms fall out of it that the rest of this article leans on. A **turn** is one round of the agent's loop: one model reply plus the tool calls it makes, so `--max-turns 6` stops the run after six rounds. And each capped run is a **chunk**, a bounded slice of the session that stops cleanly with its transcript intact. The loop works right up until the process holding it dies. And when that process dies, exactly one thing survives: the transcript. Everything else was memory. The current session ID. The retry count. The instruction someone typed an hour ago. The bare fact that a job was running at all.

An agent run is exactly the kind of job you cannot afford to forget. It runs for hours and bills by the token, it carries context a fresh start cannot get back (the explored codebase, the failed approaches, the half-written fix), and because it runs unattended against real systems, it fails at two in the morning, in exactly the state you least want to reconstruct from logs. Try to remember all of this yourself and the shopping list writes itself: durable state, a single-writer lock, a retry taxonomy, a liveness probe, a status endpoint. Somewhere around the third item you are hand-building a distributed system you never meant to own. (A session ID in a SQLite row under a process supervisor buys the first item cheaply; it is the rest of the list that compounds.) Or you can give the job the one thing the conversation already has: a record that outlives the process.

The Missing Half Has a Name

Strip the agent out of that hand-built list and what remains (durable state, one writer, retries, liveness, visibility) is not an AI problem at all. It is the standard checklist for any long job that must outlive its own hardware, and the industry has a name for the answer: **durable execution**. Temporal, the engine we used, grew out of a system built at Uber for exactly these long, crash-prone jobs.⁵

The core move is a refusal to trust process memory. A database does not keep your data safe by keeping its process alive; it writes every change to a log and rebuilds after a crash by replaying it. Durable execution does the same for *program control flow*: every step a job takes is appended to a ledger on the server, the **event history**, and when the process dies, a new one reads the ledger back and continues from exactly where the job was.



Durable Execution. The ledger, not the process, is where the job lives: any other worker replays it and picks up mid-job. No memory snapshot required.

The reconstruction is **deterministic replay**: re-run the job's code from the first line, substituting the recorded result at every step that already happened. Same code, same inputs, same decisions, same state. The catch lives in the word *same*: one clock reading, one coin flip, one network call in the replayed layer, and the replay diverges from its own history.

That constraint forces a clean split, and four plain words carry the rest of this article:

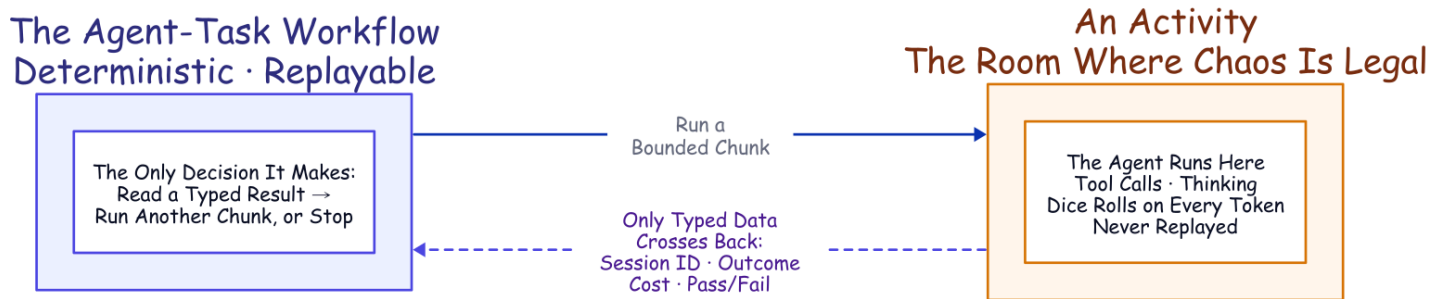
- A **workflow** is the code that decides what a job does next: the deterministic, replayable layer, the part that must never flip a coin.
- A **worker** is an ordinary process on some machine that physically does the work, and it is meant to be mortal; that is the whole trick.
- An **activity** is a single step the engine treats as a sealed box: it may call the network, write files, roll dice, anything, because its insides are never replayed, and it retries under a policy you declare rather than code.
- A **heartbeat** is a small liveness pulse a running step sends, and it may carry a little data with it. If the pulses stop for too long, the server declares that attempt dead and starts a retry that can read the last pulse the dead attempt sent.⁶

That last capability is the mechanism the rest of this design rests on.

A Room Where Chaos Is Legal

Now the plan hits the part that looks fatal. A workflow has to be deterministic, and an AI coding agent is the least deterministic software you will ever run: same prompt, different transcript, every time. Put the agent inside workflow code and the first replay diverges on the first token. That is a category error, not a knob you can tune; for a moment the whole idea looks dead on arrival.

But durable execution already has a room where non-determinism is not just tolerated but expected: the activity. Seal the entire agent inside an activity and the workflow never sees the chaos. All it sees is what crosses back as plain data (a session ID, an outcome, a dollar cost, a validated pass/fail report), and its own logic collapses to something a database could run: read a typed result, decide *continue or stop*, repeat.



The Seam. The workflow stays replayable because the agent is sealed inside an activity; only typed data crosses the line.

Two details make resume survive that boundary. First, each run gets one stable working directory derived from the job's identity, so every attempt lands in the same folder, where the transcript and the half-edited working tree both live: a retry picks up the conversation and the files it describes. Second, the workflow always chains forward the latest session ID an activity hands back, because a resume can mint a fresh one. The job's ledger stores a *pointer* into the conversation's transcript, and that pointer is very nearly the entire integration. So how does a pointer survive a murder?

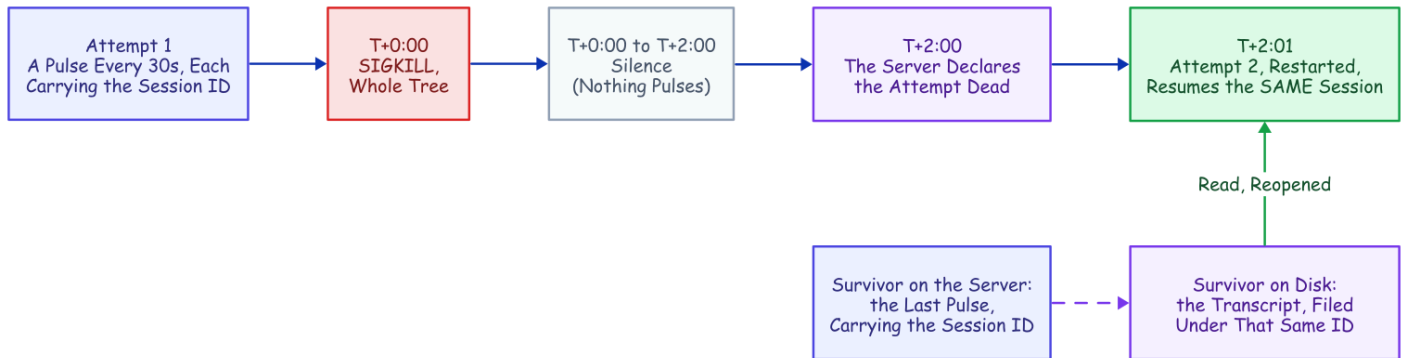
Two Silent Minutes

This is where the sealed box earns its place, and the whole mechanism fits on a clock. Here is the kill from the top of this article again, frame by frame.

1. **Every thirty seconds, all along**, a timer inside the activity has been pulsing the server: *still alive, and the session ID is 698c...*. The timer is deliberately dumb: it fires on the clock, not on the agent's output, so a long silent tool call (a slow test suite, a dependency install) cannot be mistaken for death.
2. **T+0:00**. Kill the worker and the pulses stop. Nothing was returned; no result was written.
3. **T+2:00**. After two silent minutes, the server declares the attempt dead and schedules a retry on a live worker (in this demo the same worker, restarted, because the transcript lives on that machine's disk; the sticky-queue default near the end of this article makes that landing deliberate).

4. **T+2:01.** The retry reads the dead attempt's last pulse, takes the session ID out of it, and relaunches the agent with a resume. The agent reopens its transcript and picks up mid-thought.

Step 4 works because the kill could not reach the two things that matter. Process memory died with the worker, but the transcript lives on disk, the last heartbeat lives on the server, and the session ID in that pulse is the *name* of the transcript. One link, and it is the entire recovery.



Two Silent Minutes. The last heartbeat is the checkpoint: one thin line does the entire rescue.

No completed checkpoint is required; the last heartbeat is the checkpoint. Every durable-execution engine can resume a *completed* step by replaying its recorded result; this resumes a *live* agent from an attempt that *returned* nothing at all. The unit of recovery is not a finished step; it is a coding agent's conversation, caught mid-thought.

And here, at last, is the receipt promised on the first page, one line the recovered run left in its workspace:

```
{"event": "resume_session_from_heartbeat", "attempt": 2,  
  "input_session_id": null, "heartbeat_session_id": "698c432a-..."}
```

The input session ID is null: no completed chunk existed to hand back, so the ID came *only* from the heartbeat. The run finished as the **same** session in one chunk, at \$0.0404 total: there are the four cents, and they are not a recovery penalty, just the resumed session re-reading its own context at the cache rate (the API bills tokens it has already seen at roughly a tenth of the fresh rate) and finishing the work.⁷

Two Failures, One Recovery

There are two different ways a run like this breaks, and we killed both on purpose to be sure both recover.

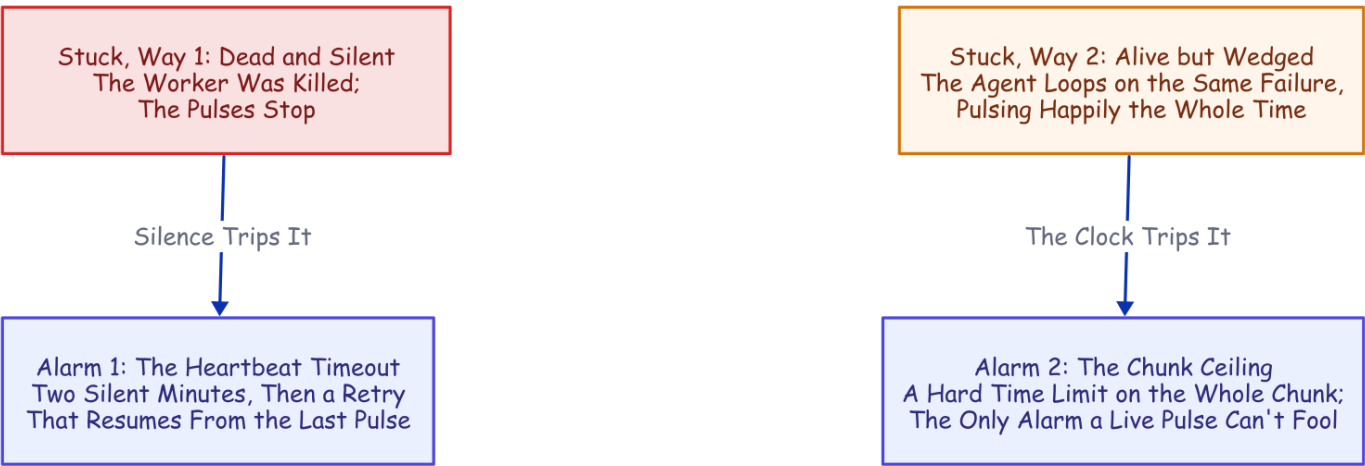
The first is the one you just watched, and it is the hard one: **the whole worker dies**. Its memory dies with it, so a *fresh* worker has to rebuild the job from the durable event history and resume the session from the last heartbeat. That is the rare, catastrophic failure, a reboot, a deploy, an out-of-memory kill, and it is the one durable execution exists for.

The second is smaller and far more common, so it earns its own test: **the agent process alone dies** while its worker keeps running. We SIGKILLED the Claude child outright, mid-chunk. Nothing global was lost, so the worker caught the broken stream, retried the chunk, and resumed the same session from the last heartbeat within seconds (the retry's first backoff), the worker itself never restarting and anything else it was running untouched.⁸



Two Failures, One Recovery. A dead worker and a dead agent process reach the same move: resume the same session from the last heartbeat. Only detection speed and worker survival differ, and we killed both on purpose to prove it.

A silent worker is not the only way to be stuck, which is why there are two alarms, not one. The heartbeat timeout above catches silence; a wedged agent needs the other alarm, because it loops forever and pulses the whole time, so the heartbeat never fires, and only a longer, hard ceiling on the whole chunk stops it.



Two Ways to Be Stuck, Two Alarms. Silence trips the heartbeat timeout; a healthy pulse with no progress runs into the chunk ceiling.

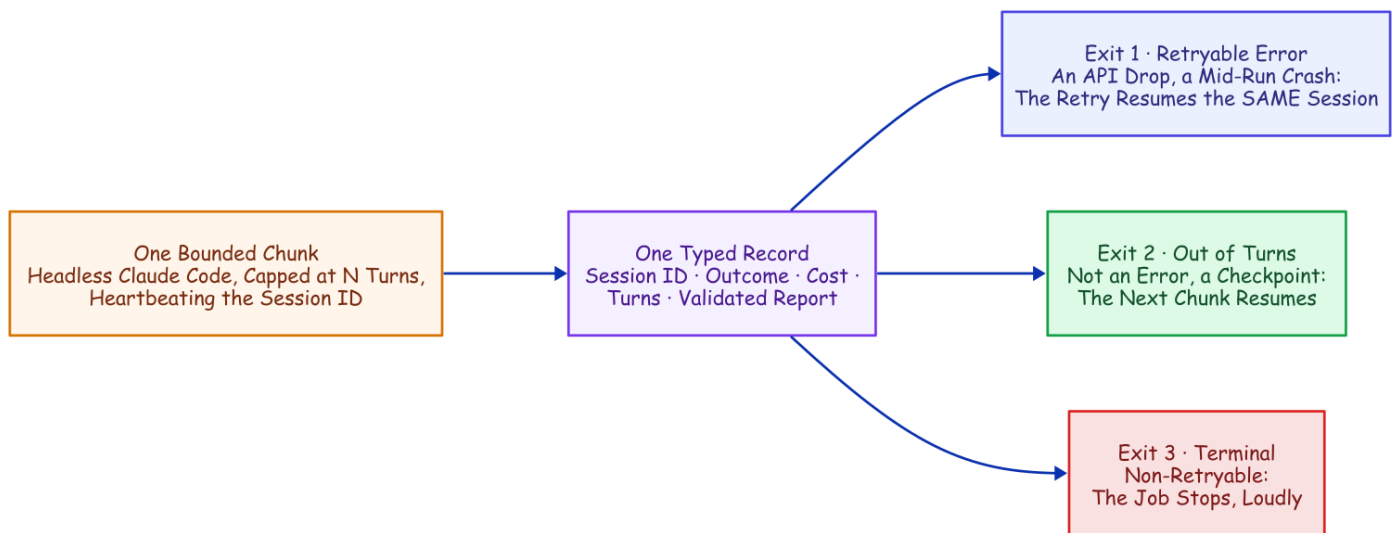
The everyday value is not that deliberate kill; a worker rarely dies outright. What actually stops a headless run is the API on the other end: an overload during a busy hour, a rate limit, a dropped stream. The agent surfaces those on its result; the activity raises them as typed, retryable failures; the retry policy backs off (five

seconds, doubling to a two-minute cap, six attempts), and every retry *resumes* the conversation instead of restarting the task. An overload window becomes added latency rather than a failed run: the same fast path the killed agent process took above, reached far more often by an API error than by a crash.

Every failure, from a rate limit to a dead worker, becomes the same cheap operation: read the last known session ID and resume the conversation, never restart the task. All of it happens inside the sealed box you watched recover, so it is worth opening that box to see exactly how one chunk is built.

Inside the Box

Open the box and one chunk is a single ordinary function. The workflow calls it, the function runs the agent, and what comes back is a typed record (a session ID, an outcome, a cost, a turn count, the validated report) and nothing else. That typing is the wall: the workflow can only read fields on that record, so nothing the agent did unpredictably can leak into the replayable layer. And the record leaves the workflow exactly three exits, which are nearly the whole control logic: a retryable failure resumes the same session under the declared retry policy; out of turns schedules the next chunk; anything genuinely terminal stops the job, loudly.



How a chunk runs. The workflow never sees inside the box; it reads one typed record and switches on three exits. Continue, resume, or stop: outcomes a database could run.

Launching the agent is a short list of settings: configuration from the project directory only, so the rules ride inside the workspace and the same agent runs identically on every worker; resume the prior session; cap the turns; auto-accept file edits, with every other tool behind an explicit allow-list; and a closing report that must match a schema, so the workflow reads a real pass/fail value instead of parsing prose.

Read the deny rules as architecture, not only safety. Deleting a tree and escalating privileges are the obvious entries; `git push` is the interesting one: the agent writes code but can never reach a remote, so the harness (this project's own Temporal-side code, distinct from Claude Code's built-in guardrails) does every push and

merge, in its own step, with its own credentials. Hooks keep the books (every tool call lands in an audit log), and the workspace re-stamps the whole policy before every chunk, byte-for-byte identical on every attempt. The guardrail cannot drift.

None of this is a bespoke protocol. It is a stock Claude Code project (settings, hooks, skills, memory), assembled fresh every chunk; the bolt-by-bolt detail lives in the engineering companion, [How It's Built](#). This shape was not our first answer. The first answer is worth showing precisely because it *worked*, until measurement talked us out of it.

The Wrong Design That Worked

An activity's result is all-or-nothing: nothing it learned reaches the history until it returns. So the first design made every turn-cap boundary a **savepoint**: finish a capped chunk, and its progress, cost, and session ID are written into the history. It worked; a savepoint-era kill test recovered on a restarted worker and finished all ten of its tests, one session end to end. But once heartbeat recovery existed, those completed-boundary checkpoints were buying a durability we already had, and each one was another resume. (The full detour, savepoint diagram and all, is in [How It's Built](#).) Worse, they were quietly doing something to the cost we did not see coming.

Three Numbers and a Slot Machine

So we measured the whole bill, holding one task constant across every scenario on the small fast model, observing everything rather than modeling it, strictly one run at a time. Three numbers carry the result.⁹

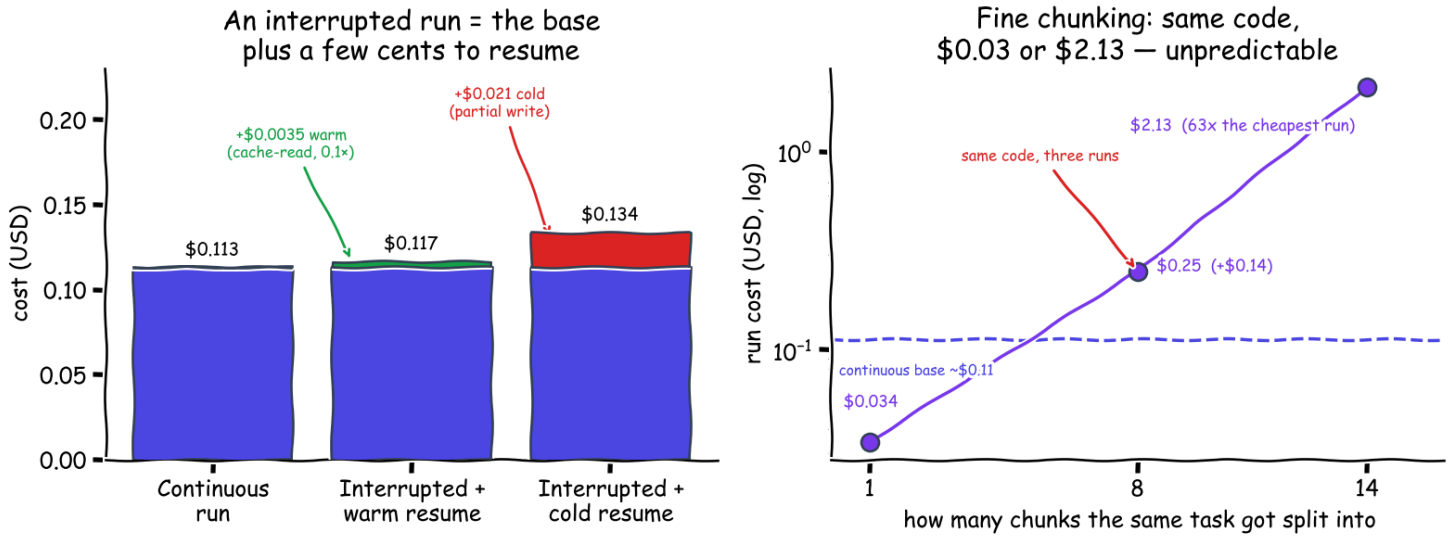
The first is the baseline. A continuous, uninterrupted run costs about eleven cents: \$0.058, \$0.132, and \$0.149 across three runs, a \$0.113 mean over an honest spread, with the agent taking anywhere from three to nine turns for the identical task.

The second is the price of surviving a crash, and it is almost insultingly small: about a third of a cent warm, two cents cold (warm meaning the cache is still live; cold meaning it expired and had to be partly re-written). A resumed session redoes no work; it only re-reads the accumulated conversation at the discounted cache rate. Even a cold resume, after the session sat idle for over an hour, paid only a partial cache re-write and then re-warmed.

The third number is the warning. The same task, chopped into fine two-turn chunks on identical code, ran \$0.034, then \$0.25, then \$2.13: a sixty-three-fold spread, with the worst run costing nineteen times the continuous base. And the culprit is not the mechanism we expected. The per-boundary cache re-read stays a cheap fraction of a cent; the culprit is **behavioral**. A tight turn cap changes what the model does with its turns, and a run that wanders across fourteen chunks and twenty-eight turns (to finish what a continuous session did in nine) pays for every wander. (An earlier, smaller experiment had even sworn chunking was cost-neutral; the fuller one shattered that on a more open-ended task. The correction story, the method, and

the pricing canary that re-checks every number on a schedule are the economics companion, *Mechanics Cost Cents, Behavior Costs Dollars.*)

Same TDD task, haiku, measured: a resume is cheap; fine chunking is unpredictable



What Durability Costs (measured, small model). Left: a warm resume only re-reads context at the cache rate; a cold one pays a partial re-write. Right: fine chunking adds near-zero to two dollars, unpredictably.

So the rule is large chunks by default: fine chunking is the only lever that meaningfully moves the total, and it moves it unpredictably. And notice what every number in this section has in common, because the pattern recurs across everything this project measured: **the mechanics cost cents; the behavior costs dollars.** Small chunks, though, earn their keep for a different job entirely, which is next.

Query, Steer, Cancel: What the Boundaries Became

Completed-chunk boundaries did not stop being useful when heartbeats took over the durability. Relieved of the safety job, they were reborn as the *visibility and steering* cadence: the moments where a running job can answer to verbs a bare process cannot.

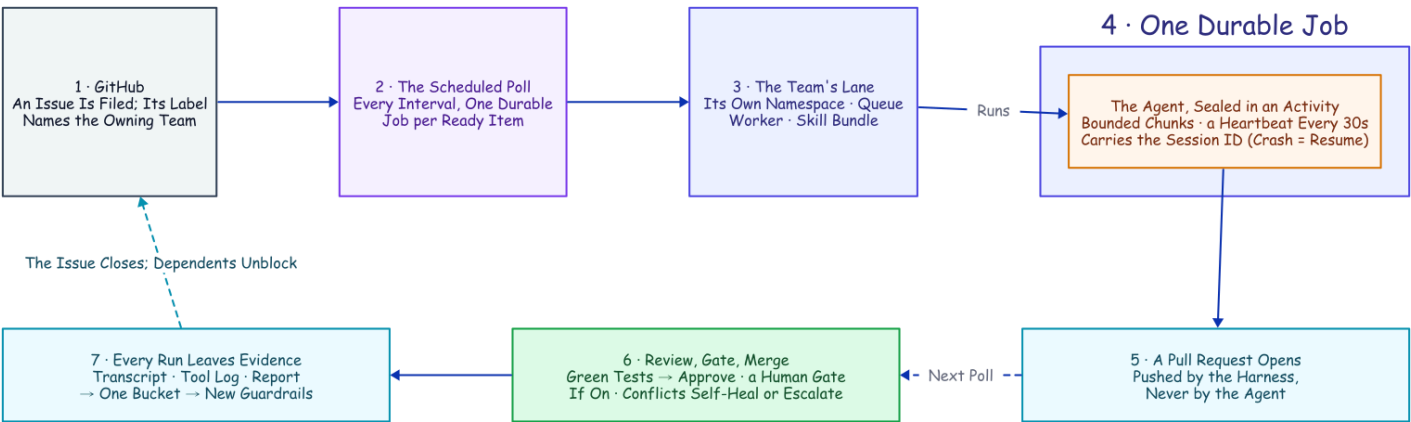
Query it. A read-only question, answered from the workflow's own state rather than from logs. Mid-run, a status query comes back with the chunks completed so far, the running cost, and the live session ID. **Steer it.** A **signal** is Temporal's message to a running workflow, no relation to the Unix signal that killed the worker earlier. One injected *"also add a stats subcommand, same test-driven treatment"* into a running job, and the workflow folded the instruction into the next chunk's prompt. And if a steer arrives during the very last chunk, it triggers one more, so late guidance is never silently dropped. (That demo is recorded in the lab notebook, the running experiment log this project distills.) **Cancel it.** Delivered on the next heartbeat: the activity tells the running agent to stop, so it exits cleanly instead of burning tokens as an orphan, and the job lands in a canceled state you can see.

So the turn cap has one clear rule: it sets how often the job reports in and can be redirected. Make it small only when a human needs to watch or steer in near-real time. And the left column of this table is the hand-built shopping list from the start of this article, paid off item by item:

What you would build by hand	The primitive that already exists
A session-ID file plus a lockfile	Job-ID uniqueness: one job is the session's single writer
A restart script plus a liveness probe	A heartbeat timeout plus a retry policy, declared not coded
A task-to-session map in memory and on disk	The session ID in heartbeat details mid-run, and in workflow state after a chunk
A hand-rolled taxonomy of which errors to retry	Typed retryable vs. non-retryable failures, with backoff
A status endpoint scraped from logs	A progress query plus the event-history UI
Steering a running task (never actually built)	A signal, folded into the next chunk's prompt

Point that same machinery at a whole repository and it does something bigger. The heartbeat, the resume, and the declared retries that just carried one agent through a crash are what will carry a whole *team* of single-purpose durable jobs through the issue-to-merge run promised at the top, conflict and all, only a few of them ever calling the agent. That team is the rest of this article.

Nine Specialists, Two Conductors



The whole system, one loop. Hold this picture; the rest of this article, and every companion in the family, zooms into one box of it.

Everything so far makes *one* job durable: the one that runs the agent. A durable job, unlike a bare process, can be given an owner, an address, and rules. So the team is not that agent cloned into a crowd; it is more single-purpose durable jobs built around it, each with its own mandate, every one inheriting the same crash recovery, the same resume-from-heartbeat, the same declared retries. A nine-member team is nine durable

jobs under one protection, and only the few that call a model are agents at all. (A lane, a term defined in a moment, is one team's own slice of the system.)

Here is what we actually built. **Two workflows conduct the delivery team.** One is the durable task you have already met: it runs the agent in bounded chunks and then ships the result. The other is a scheduled poll that looks at a repository for new work, on a timer that is always counting down toward its next sweep, as a durable job in its own right; keep that timer in mind, because it quietly drives everything that follows. And beneath those two conductors sits a team of single-purpose activities (nine in the runs measured here; twelve today), **each with exactly one job.** None of them is clever. Each is a sealed box that does one real-world thing and hands back typed data, and that narrowness is the point: it is what lets the retries, the heartbeats, and the audit trail apply to every one of them for free.

Nine specialists, two conductors, and a filed issue can travel the entire distance to a merged pull request while the humans sleep. Name the team by what each member does. One activity **runs the agent**: a single bounded, heartbeated chunk, the only member that runs a model. One **exports the transcript** into a readable audit log once the work is done. One **scouts the repository** for new issues and pull requests and routes each to a team. Three handle the pull-request lifecycle: one **opens the pull request**, one **posts the review verdict** (approving only if the tests pass), one **merges** the approved change. And three handle the case where a merge fights back: one **updates a stale branch** by merging in the base and retrying, one **escalates a real conflict** to the team that owns the code, and one **pushes the resolved branch** so the merge can finally land. The shape should feel familiar: it is the discipline of a hardened CI/CD pipeline (gated merges, least-privilege credentials, an audit log on every step) applied to a team whose members are durable jobs.

The reason this deserves the word *design* is that adding to the team is mechanical. Want a new capability (a security scan, a changelog write, a deploy)? You write one function that does the real-world thing and hands back typed data, you declare how it should retry, and you name the moment in the workflow where it runs. It inherits crash recovery, backoff, cancellation, and the audit trail from the harness, not from anything you wrote. That is the whole shape of "write the happy path and rent the rest."

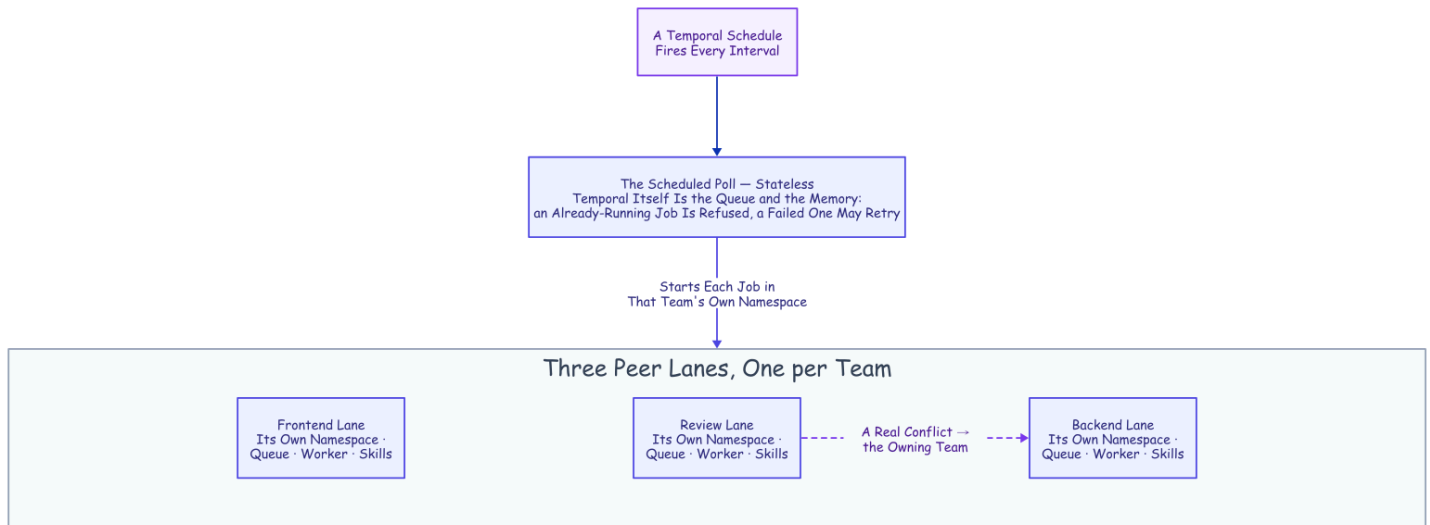
Trusted by the Queue, Not the Prompt

The team scales by giving each *kind* of work its own lane, and every "team" noun here is a concrete primitive underneath. A **namespace** is an isolated partition of the Temporal server. A **lane** is this article's word for one team's namespace plus everything the team owns inside it: its own work queue, and its own bundle of skills bound into the workspace before each chunk runs. The teams in these runs were backend, frontend, and review; the system runs six lanes today. Workers listen only to the lane they own.

And this is the part worth pausing on: a backend worker is trusted to do backend work not because its prompt says *act like a backend engineer*, but because it listens on the backend queue, runs under the backend team's committed playbooks, emits backend audit artifacts, and answers to the same retry, cancel, query, and cost machinery as every other lane. **It is trusted by the queue it polls, not by the prompt it was handed.**

The skill bundle is a contract the lane owns (test-first, review-your-own-diff, the required report format), and the load-bearing rule inside it says a change in one layer has to prove it did not break the others before review will approve it.

The formula for a team member, then, is short. **A mandate:** the one job it owns. **Skills:** the playbooks for how it operates. **Hooks:** the rules enforced on every tool call. **And a settings file stamped from human-committed harness code.** All four ride in version control; none of them lives in the model. You could swap the model tomorrow and the team would still be the team.

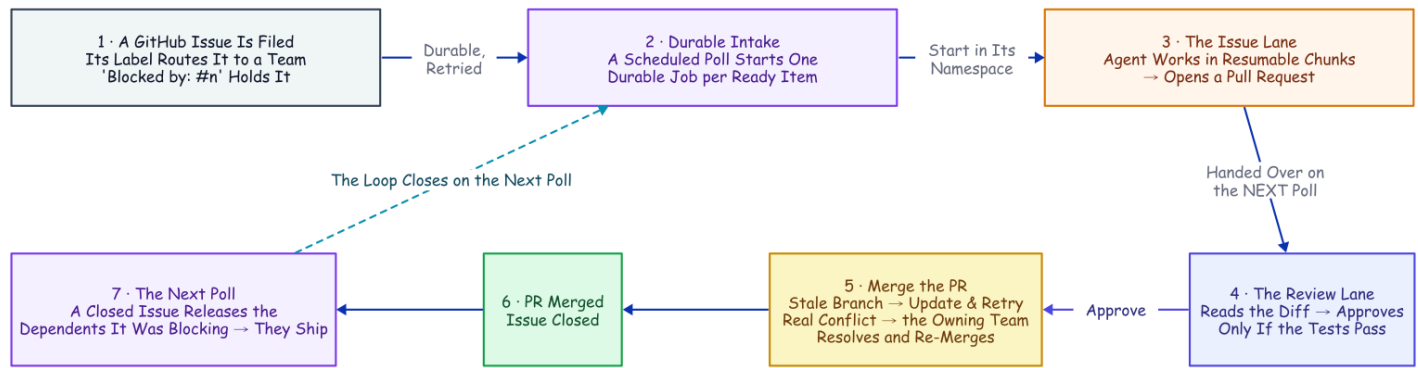


Trusted by the Queue, Not the Prompt. Three identical peer lanes; only the queue they poll differs. The scheduled poll and the cross-lane escalation both reach another namespace the same small way: a client call made from inside an activity.

One more primitive and one structural catch finish the picture. **One owner per piece of work** comes from deriving each job's ID deterministically from the issue and starting it with an "allow duplicates only after failure" policy: a second attempt at a running or completed job is rejected by the server, while a failed job may retry on a later sweep. That single fact is *why* the poller can be completely stateless (re-seeing the same issue is a no-op the server refuses, so the poll keeps no memory of its own), and why a transient failure heals itself instead of deadlocking an issue forever. And one structural catch shapes everything. A workflow can only start child workflows inside its own namespace. So when the intake or an escalation has to cross a lane, it goes through an activity instead: an activity may do arbitrary I/O, so it opens a client to the target namespace and starts the job there. Remember that move; you are about to see it twice.

One Issue Walks the Whole Loop

Put the conductors and the team together and one issue travels a full loop, every box along the way a durable step in the history.



From Filed Issue to Merged PR. Every box past the filed issue is a real activity from the team. The conflict detour and the unblock loop are not special cases; they are the same durable machinery running one more time.

Intake. A schedule fires the poll on an interval. The poll reads the repository (open issues, open pull requests, which issues have closed), routes each item to a team by its label, and starts one durable job per ready item with a deterministic ID: the first of the two client-from-inside-an-activity crossings. An issue marked *blocked by* another is held until that other issue closes.

Build. The issue's job runs the agent in bounded, resumable chunks, heartbeating its session ID the whole time; the agent commits as it works, so the branch accumulates a real history of checkpoints rather than one anonymous blob. When the agent reports success, the transcript is exported to a readable audit log and a pull request is opened from the work branch, carrying that full commit history, pushed by the harness, never by the agent.

Review and merge. The open pull request is picked up on the next poll and routed to the review lane as its own job. The review agent reads the diff and runs the suite, and its verdict is a single field in the required report: a pass/fail boolean that *is* the merge switch. True posts an approval and merges; false requests changes, and in the measured runs the trail ended there: the bounded fix loop that now routes a denial back into a fresh build round arrived later (*The Human Is a Durable Object* tells it). A merge against a branch that has fallen behind triggers an automatic branch update (the same "update this branch" move you would click in the GitHub UI) and a retry.

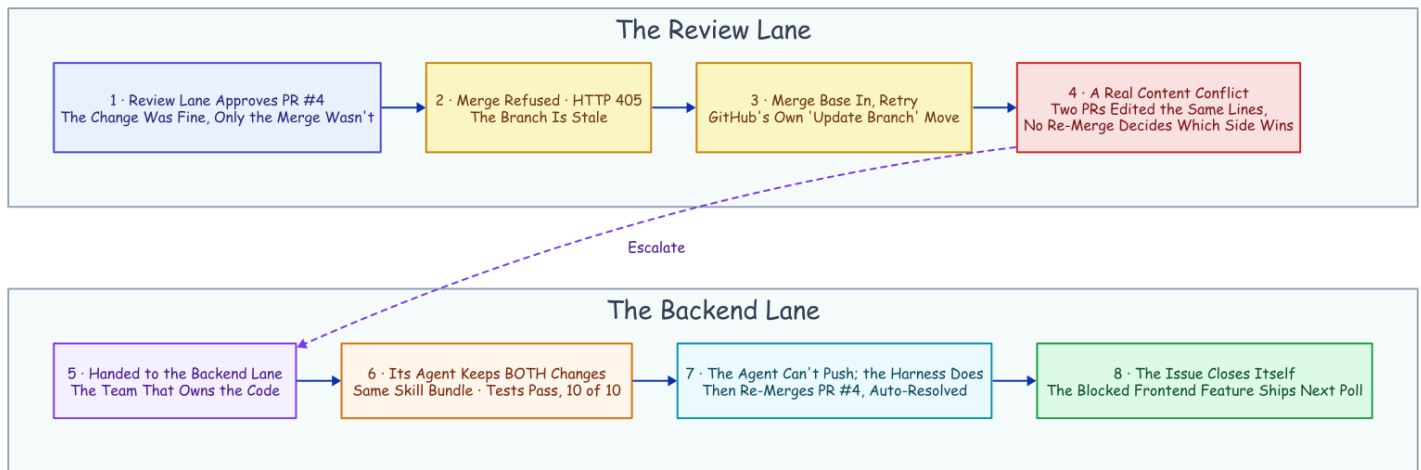
Escalate, resolve, unblock. A retry that *still* conflicts is a genuine content collision, and it gets handed to the team that owns the code: a resolve job in that team's own namespace, where an agent fixes the conflict by hand, runs the tests, and the harness pushes and lands the merge. When that merge closes the issue, the next poll releases whatever was blocked on it, and the chain runs again for the dependent. No human decision anywhere in it.

That is the happy path, walked once. What proves the team is real is the day the path broke.

The Conflict That Resolved Itself

Every piece of that machinery was about to be needed at once, so we built a real target for it: a full-stack "snippets" web app, backend and frontend and browser tests, on a live deployment with a namespace per team and the poller on a schedule.

Two backend issues landed in the same poll window and collided by construction. One added search; the other added a delete endpoint. Both edited the same regions of the same backend file and its API contract. Each issue got its own branch, its own job, and its own pull request. The search PR merged first. And the moment it did, the delete PR (**PR #4**, the one this story follows) went stale: main had moved underneath it. This is where autonomy usually ends quietly: a "needs a manual rebase" comment, and a human inheriting the mess in the morning.



The Conflict That Resolved Itself. Detection lives in the review lane; the fix lives in the lane that owns the code.

Here is what happened instead, every step a durable activity in the history.

The review lane approved PR #4 on its merits. The code was fine; nothing about the *change* was wrong. But the *merge* was refused, because the platform will not merge a stale branch.

The workflow tried the boring fix first. Merge main into the branch and retry, the same "update branch" move from the pipeline above. That is when the real problem surfaced: the two PRs had edited the same lines, so the update produced a **content conflict**. No amount of re-merging decides which side wins; someone has to read both changes and write the version that keeps both. That is a judgment call, and judgment is what the agent is for.

So the review lane handed the conflict to the team that owns the code. It read the issue number out of the branch name, looked up that issue's team, backend, and started the resolve job **in the backend namespace, on the backend queue**: the same client-from-inside-an-activity call the scheduled poll uses, the move you have already seen. The agent that picked it up was nothing special: same skill bundle, same policy, same durable harness; only the prompt was different. It merged main into the branch, kept both the search feature and the delete endpoint, and ran the tests on the combined result: ten of ten green. Denied `git`

push like every agent here, it left the pushing to the harness, which pushed the branch and merged it. "Merge PR #4 (conflict auto-resolved)." The issue closed itself a moment later.

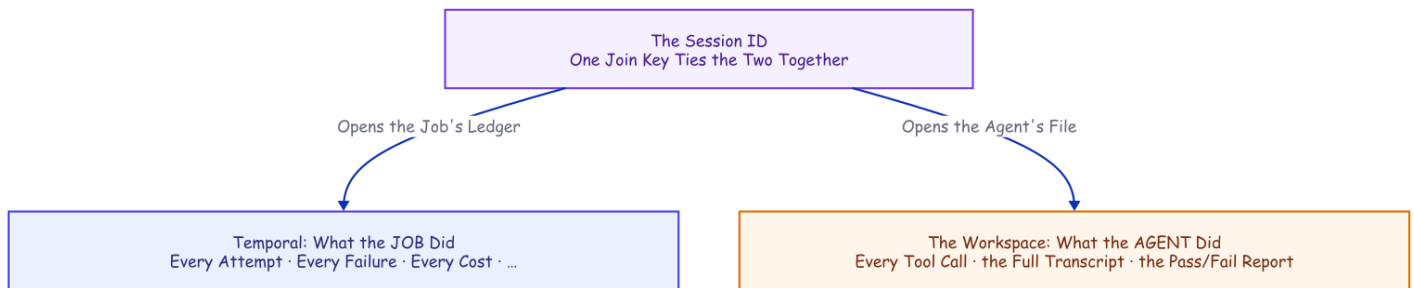
Two properties keep this from being a party trick. The escalation cannot loop: the resolve job's ID is derived from the PR and its current head commit, so a duplicate start is refused by the server and cascading conflicts converge instead of multiplying. And it cannot cheat: the escalation is just a workflow starting another workflow, under the same crash recovery and hard limits as everything else.

Then the part that turns an incident into a system. A frontend issue had been sitting in the queue the whole time, marked *blocked by* the backend issue that just closed. The next poll noticed and released it. The frontend agent built a delete button against the very endpoint the resolution had just landed, added a browser test, and opened a pull request the review lane merged. From collision to shipped feature, the chain ran without a human decision in it, one mechanical asterisk included.¹⁰

A claim that size earns a question: how would you even know it is true? You would read the record, and every run leaves one.

Every Run Leaves Evidence, and the Evidence Improves the Team

Because the agent runs under a policy the workspace owns, every run leaves *two* planes of evidence, joined on one key: the session ID, the same ID that rode the heartbeat on the first page. Temporal owns the record of the **job**: every activity, attempt, failure, heartbeat timeout, and recorded cost, queryable long after the run is over. The workspace owns the record of the **hands**: a hook appends every tool call to a log as it happens, and a closing activity exports the whole conversation to readable Markdown filed under the session ID.



Every Run Leaves Evidence. Ask what the job did and you read Temporal's history; ask what the agent did and you read the workspace's package. One session ID opens both records.

Collect those exports and you have a corpus of how your agents actually behave, shipped to a bucket, one folder per run. And a corpus is a feedback loop waiting to be closed, so we closed it. A reviewer read the tool-call logs across the cost-experiment runs and found one consistent waste: the agent kept re-checking work it had already done, listing a directory to confirm a file it had just written, re-running tests that were already green. We codified the lesson two ways, a written rule and a deterministic hook, and re-ran. The rule bought a modest ~20% drop (half an instance per run, inside run-to-run noise) and one run ignored it entirely; the

hook flagged every remaining instance, twelve out of twelve. **A line in a prompt is a suggestion; a hook is a law.** And a mined corpus is how the laws get written. The loop has held since: on the latest overnight run of the full team, the audit trail those hooks write was the record that debugged five live failures into five same-evening commits, one of them the dedup-deadlock fix you will meet in the defaults below. The system's job is not to avoid failure; it is to convert failure into a commit.

What This Doesn't Solve

Five real gaps, told plainly, because the honest ones are the load-bearing ones.

- **The filesystem is not checkpointed, and steps run at-least-once.** A retried chunk resumes against a directory the dead attempt half-mutated. It converged in every test here, but *converges in practice* is not *idempotent by construction*: outward actions want idempotency keys, and the real fix is a fresh git worktree per chunk. The transcript is the same uncheckpointed local file, but a torn last line proved benign: truncating a session's final JSONL entry, or appending a partial one, still resumed with full context both times (measured, in a staged torn-transcript run).
- **A permanently dead machine takes the workspace with it.** The transcript and working directory are local files; a host that never returns loses them. Restarts are covered by sticky-queue pinning below; true machine loss wants shared storage. A two-filesystem run has since proven the resume (machine A's disk deleted, machine B finishing the same session from the heartbeat ID; [How It's Built](#) has the receipt). The harness still does not ship the sync itself.
- **A hard kill can orphan the agent.** The child process may finish its chunk anyway, racing the retry: the effect that masked our first recovery attempts. Production workers belong in a process group that dies with its children. The same race needs no kill at all: a paused or partitioned worker looks dead while its chunk runs on, so the retry can overlap it. Sticky pinning narrows that to one host; a fencing check at chunk start would close it, and we have not built one.
- **Every input is trusted.** Issue text, PR bodies, and code comments are all prompts, and here only the author writes them. A public deployment is a prompt-injection surface at each one; the mitigations (author allow-lists, injection-aware review, the human gate) are not built here.
- **The merge gate is an agent's verdict.** Approve-and-merge turns on one schema-validated boolean: the review agent's tests-pass verdict. A companion measured that self-report against ground truth: wrong three in ten, every miss a false alarm, all from the builder arm; the reviewer arm was honest five for five, but only on red work it could not touch, so the green-work cell a real merge rides on is still unmeasured (*The Agent Grades Its Own Homework*). For this demo that is the point; for production, a placeholder. Put your human gate there: a protected branch, or the durable approval step proven live in *The Human Is a Durable Object*.

Defaults Worth Stealing

If you build any version of this, on any stack, these are the settled defaults this project would hand you. Each one was paid for above.

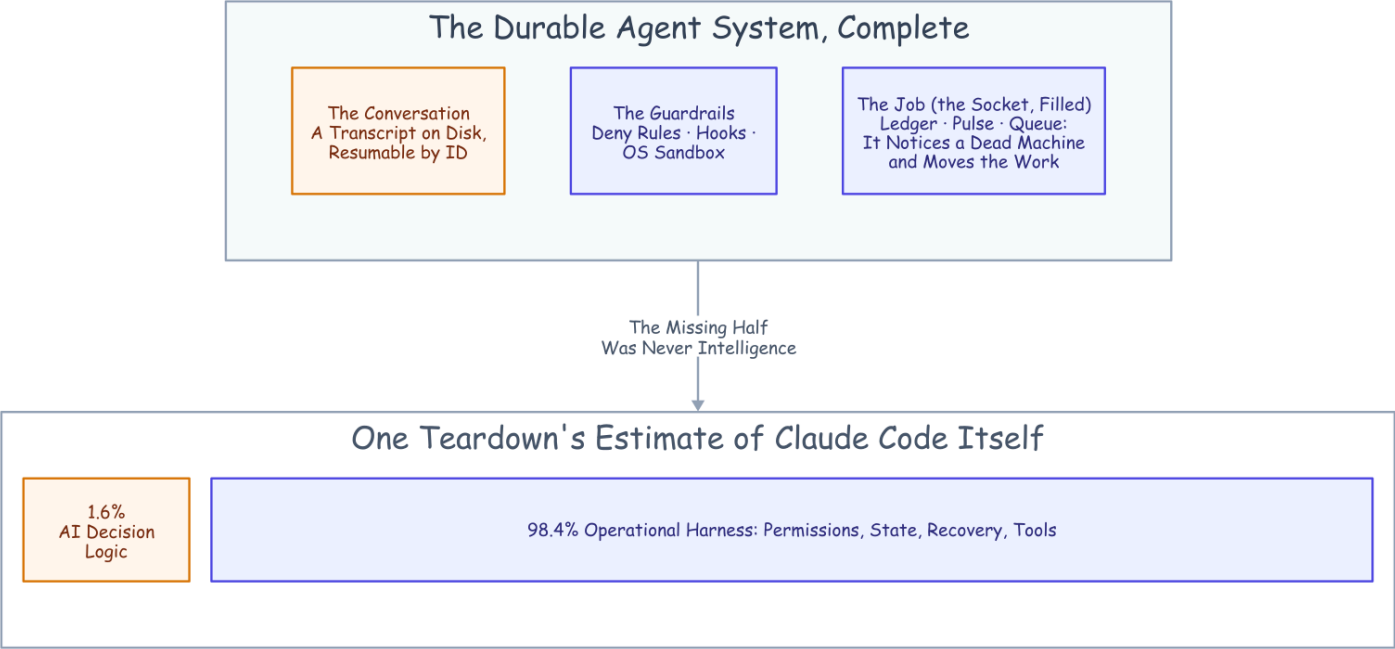
- **Big chunks by default.** A tight turn cap is a slot machine: the same task ran \$0.034 to \$2.13 on identical code. Chunk for visibility and steering, never for durability.
- **Heartbeat a resume handle, not just liveness.** The last pulse is a free checkpoint; a retry that can read it never starts from zero.
- **Kill, and deploy, by process group.** An orphaned agent child finishes the work anyway and lies to you about whether your recovery works.
- **Deny git push to every agent.** The agent writes code and commits as it works (commits are local, a ledger inside its own workspace), but it can never reach the remote. When work needs pushing or merging, a separate harness activity does it, with its own credentials, as a recorded step in the durable history, carrying the agent's full commit history with it. No agent holds a world-changing credential, and every outward change has an audit entry.
- **Route by queue, not by prompt.** A lane's queue and skill bundle are an identity a worker cannot drift out of; a persona in a prompt is not.
- **Pin the resume to the worker that holds the session.** Opt-in sticky queues: the first chunk runs on the shared lane queue and reports its worker's own stable queue; the workflow pins every later chunk there. A restarted host resumes correctly, and a chunk never silently lands where the session is absent.
- **Deterministic job IDs, duplicates allowed only after failure.** Running and completed jobs are refused, so the poller needs no memory and the escalation loop cannot run away. Failed jobs may retry on a later sweep, so one transient error cannot deadlock an issue forever. (A live fleet run found the stricter refuse-everything version of this rule doing exactly that; the evidence archive has the receipt.)
- **Write the rule as a hook, not a prompt.** Measured head-to-head, the rule was always probabilistic and the hook always law, down to a rule losing outright to a direct instruction while the hook caught twelve of twelve. Priced at the tool boundary in *Flag, Block, or Beg* and at the finish line in *Done Is Not a Claim*.

The Harness Was the Hard Part All Along

One practitioner teardown of Claude Code's own codebase estimates that roughly **1.6%** of it is the AI decision logic. The other **98.4%** is operational harness: permissions, state, recovery, tools.¹¹ Set that next to the hand-built list from the start of this article (and the table that mapped it, item by item, to existing primitives) and the overlap is hard to miss. The harness every agent team is rebuilding is, almost line for line, what durable-execution engines have provided for a decade.

The two sides are already circling each other. Temporal's own AI cookbook wraps model *API calls* in activities; a production write-up wraps the agent framework the same stateless way; another project built durable checkpointing around whole Claude Code runs on its own runtime.¹²¹³¹⁴ In the sources we checked, none composes the two the way this project does: headless Claude Code *sessions*, the session ID carried in heartbeat details mid-chunk and in workflow state after a completed one. That claim has a short shelf life, so re-check it before you build.

Claude Code brings the durable conversation. Temporal brings the durable job. The seam between them is one pointer: a session ID, carried in a heartbeat, chained through a ledger. That is the whole composition, and it is why the kill on the first page is staged as a stunt and lands as the boring case. The agent could always remember the conversation. The merged pull requests, the conflict that resolved itself, the failures converted into commits: all of it is what became possible the moment something finally remembered the job.



The whole, at last. The missing half was never intelligence.

The Family

This article is the trunk; six companions each own one branch, and they read in this order:

- [How It's Built](#), the rivets: the settings, the tuning, the lane plumbing, the details the story above skips
- [Mechanics Cost Cents, Behavior Costs Dollars](#), the bill: every boundary priced, the method, and the canary that re-checks it on a schedule
- [Flag, Block, or Beg](#), the tool-call boundary: what a prompt, a flag, and a block each buy, measured
- [Done Is Not a Claim](#), the finish boundary: the same hard deny, moved to the exit, inverts the result
- [The Agent Grades Its Own Homework](#), the verdict boundary: the merge switch's boolean against ground truth
- [The Human Is a Durable Object](#), the person: the one human decision, modeled as durable state, deny-safe on silence

Disclaimer: the views and opinions expressed in this article are my own and do not necessarily reflect those of my employer. This is a personal project, not affiliated with or endorsed by Anthropic or Temporal.

Sources

- **Vendor documentation** for every Claude Code behavior cited (headless mode, sessions, the CI action, permissions, hooks, sandboxing) and every Temporal behavior (durable execution, activities, heartbeats and their throttling, versioning): code.claude.com/docs, docs.temporal.io, python.temporal.io; specifics footnoted inline. *Vendor/canonical*. Checked 2026-08-20.
- **Measured runs**. The heartbeat recovery, the conflict run, and the learn-loop are recorded in the project's evidence archive: each has a results file there, with its runner and analyzers beside it. Agent model `haiku`, July and August 2026. The cost measurements live with the economics companion, *Mechanics Cost Cents, Behavior Costs Dollars*.
- **Practitioner and adjacent sources** (the 1.6%/98.4% teardown, the Temporal AI cookbook, and the two adjacent durable-agent write-ups) are footnoted inline.

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1. **Claude Code headless mode**: `--output-format json` returns `session_id`, cost, and (with a JSON schema) structured output; `--resume <id>` continues a stored session; transcripts are stored under `~/.claude/projects/<cwd-slug>/<session-id>.jsonl`, the working-directory path with non-alphanumerics replaced. Checked 2026-08-20. *Vendor/canonical*. ↩
 2. **[anthropics/claude-code-action@v1](#)** (GA) runs headless Claude Code from GitHub workflows: it activates on `@claude` mentions, issue assignment, or whatever events the workflow declares, `pull_request` and cron included, and lists automatic PR review and turning issues into pull requests among its documented use cases; its configuration is aligned with the Claude Code SDK. Checked 2026-08-20. *Vendor/canonical*. ↩
 3. **Claude Code permissions** are evaluated deny → ask → allow (a deny is unoverridable) and "enforced by Claude Code, not by the model"; **PreToolUse** hooks can deny a call before it runs, while **PostToolUse** hooks receive the completed event as JSON and report back; Bash sandboxing uses Seatbelt (macOS) / bubblewrap (Linux). Checked 2026-08-20. *Vendor/canonical*. ↩
 4. **Claude Code sessions**: since CLI v2.1.223 the resume lookup searches the current project directory and its git worktrees first, then every other project on the same machine (before that release it stopped at the project directory). Either way there is no documented cross-machine session transport or leasing: the gap an external orchestrator must fill. Checked 2026-08-20. *Vendor/canonical*. ↩
 5. **Temporal docs**: durable execution persists workflow progress as an event history; workflows must be deterministic so history replay reconstructs state; activities host side effects and are retried by policy; Temporal descends from Cadence, built at Uber. *Vendor/canonical*. ↩
 6. **Temporal Python SDK**: `activity.heartbeat(*details)` sends heartbeat details, and `activity.info().heartbeat_details` exposes the previous attempt's details to a retry. The worker throttles how often details are persisted (`max_heartbeat_throttle_interval / default_heartbeat_throttle_interval`). Checked 2026-08-20. *Vendor/canonical*. ↩
 7. The recovery run, 2026-07-15: the worker's whole process tree was SIGKILLED mid-chunk before any completed result; attempt 2 recovered the heartbeat session ID (with the input session ID null) and relaunched with resume; it completed as one chunk, \$0.0404, the same session. Requires a heartbeat throttle short enough to persist the ID, and a process-group kill so the agent child can't orphan-finish the work. One accounting honesty: whatever the dead attempt burned before the kill never returned a result, so it appears in no ledger; the figure is the surviving attempt's bill. A replication on 2026-08-20, on a newer CLI, confirmed the same recovery (input session null, heartbeat ID used, same session finished) at \$0.1845: identical mechanism, more agent turns, the behavioral variance the economics companion prices. ↩

8. The child-kill runs, 2026-08-30, model `haiku`, lane `testing`: only the Claude CLI subprocess was SIGKILLED mid-chunk while its worker stayed up. The SDK surfaced the dead CLI at once, the activity raised a retryable error, and attempt 2 resumed the *same* session from heartbeat details (input session ID null) within about five to ten seconds across two runs, the retry policy's first backoff; the worker was never restarted, and it completed as one chunk (\$0.1362, then \$0.1740). This is the same resume-from-heartbeat path as a whole-worker crash, caught by a raised error instead of the two-minute liveness timeout. Runner and results file are in the evidence archive. ↩
9. The four-cost experiment, model `haiku`, 2026-07-15, run strictly sequentially, everything observed rather than modeled; the method and per-run detail are in the economics companion, *Mechanics Cost Cents, Behavior Costs Dollars*. ↩
10. The asterisk: PR #4 predated the escalation feature, so its review was re-triggered against the new code, and this run's polls were hand-triggered rather than waited out on the timer. The scheduled poll and the first-pass routing are proven across the other recorded runs; this run isolates the resolution chain. The full run record is in the evidence archive. ↩
11. *Dive into Claude Code* (arXiv 2604.14228): ~1.6% of the codebase is AI decision logic; ~98.4% operational harness. *Practitioner/academic*. ↩
12. *Temporal AI cookbook*: the "Basic agentic loop with Claude" uses model Messages API calls as activities; no Claude Code CLI, no session resume. *Vendor/canonical*. ↩
13. "Building fault-tolerant long-running AI workflows with Claude Agent SDK × Temporal.io", claudelab.net (2026-04-22): wraps API messages in activities; no headless sessions, no cross-activity resume. *Practitioner*. ↩
14. "Don't make Claude do the same work twice", zenml.io (2026-06-01): durable checkpointing of Claude Agent SDK runs on ZenML's own runtime, not Temporal. *Practitioner*. ↩